

Assignment - Update & Discussion Brief 2

When: In class on **February 24, 26 or March 2, 4**. Your specific presentation date will be assigned.

Where: 5H47

UOD: Service Dress

Length: 25 minutes (+1/-2) + 10 minutes of questions/discussion

Why?

This is your team's chance to share the story of your project with the class. Think of it as a *comprehensive* project review where you update us on your progress, but more importantly, **teach us what you've learned**.

Your goal is to explain your project, the methods you've tried, and the results you're seeing so far. This is another opportunity to generate discussion and get valuable feedback from your peers and a faculty panel. The focus of this briefing should be on your preliminary results, but it still must include all sections (see rubric below).

What to Include

Your presentation should walk us through your project from start to finish. Make sure you cover these key areas:

Your presentation should follow a clear, logical narrative that walks us through your project. The sections below are aligned 1-to-1 with the grading rubric.

- **Bookkeeping Slides:**

- **Title Slide**
- **Team Slide:** Include your sponsor(s) and mentor(s).
- **Overview:** Your agenda for the talk.

1. **Introduction & Problem Framing**

- Present a "broad brush" look at the project and its motivation.
- Present a definitive **purpose statement** that outlines the "desired end state" and "overall goals".

2. **Theory & Literature Review**

- Explain relevant **Theory** and novel methods. Assume your audience has little to no background.
- Discuss **Prior Work**, including published works and USAF reports.
- Clearly state the knowledge gap which you intend to fill.

- You **must** use published sources w/ citations & references.

3. Research Question(s) & Data

- State your specific **Research Questions**. These must be measurable, detailed, and related to the project purpose.
- Describe your **Data**. Discuss its provenance, collection, strengths, flaws, and format.

4. **Methodology & Analysis

- Detail your **Methodology** and the **Analysis** techniques you are using.
- How are you approaching data cleaning? What models will you use? How will you avoid overfitting?
- If you have used novel methods from your lit review or other sources, be sure to explain them sufficiently to the class

5. Results & Discussion

- Provide your **Preliminary Results**. How does your work so far align with your research questions?
- What do your results tell you?
- **Embrace Failed Models!** This is the section to discuss the model and its failings.
- What are your **Lessons Learned**? What have you learned about the problem, your failed models, or the Data Science process?

6. Conclusions

- What is the answer to your **Research Question**?
- What decision-quality information have you provided to your sponsor?
- Do you have any **Lessons Learned**?
- Have you answered the questions your sponsor was asking?
- In the next half-semester, how will you ensure that you have answered the **Research Question** and provided decision-quality information to your sponsor?

• References & Documentation

- **References:** Include citations throughout your presentation and include a references slide listing the full citation.
- **Acknowledgements:** Note anyone *not* on the team who contributed.
- **AI Usage:** If you used generative AI, document your usage with the 3-part documentation statement and transcript link.

A Key Point: Embrace Failed Models!

Data science is an iterative process, and "failure" is just another word for "learning." A model that doesn't work is still a result.

You are highly encouraged to **embrace failed models and results**. Explaining *why* something didn't work and what you learned from it often leads to the **most insightful discussions**, which is a major point of this brief.

Format, Audience, and Logistics

- **Format:** This is a formal group presentation in **Service Dress**.
- **Length:** You will have a strict **25-minute** time slot for your presentation. This will be followed by a **10-minute** period for Q&A from the panel and your classmates.
- **Participation:** **All group members must speak**. Plan your transitions.
- **Audience:** We have extended invitations to Department Heads, the Head of the research office, and your project sponsors.

How You Will Be Graded

This brief will be **formally graded by a faculty panel**. The panel will include:

- Maj Merrick
- Your faculty mentor
- One faculty member from DFMS
- One faculty member from DFMA
- One faculty member from DFCS

The panel will use the rubric below, which is aligned with the A/B/C/D standards from the Course Letter.

Update & Discussion Brief Rubric

**Cadet Team
Members:**

Project Title:

Evaluator Name:

Category	A	B	C	D/F
1. Introduction & Problem Framing	Clearly and concisely introduces the sponsor, background, and problem. The problem is well-motivated and its importance is established.	Introduces all required elements, but the problem framing or motivation could be more focused.	The introduction is underdeveloped or lacks focus. The problem's importance is not established.	The introduction is minimal or incomplete. It fails to define the problem or provide necessary context.
2. Theory & Literature Review	Demonstrates "high quality" independent research. Clearly explains relevant theory and "new topics" from the literature, connecting them directly to the project.	"Independent research was solid but could be more comprehensive". Explanations of new topics are present but could be more comprehensive or clearer.	"Independent research was limited or not fully developed". Fails to explain new topics or connect them to the project.	"Independent research was minimal" or "nonexistent". Provides no context from theory or literature.
3. Research Question(s) & Data	The research question is well-defined, specific, and clearly motivated. The data source, scope, and limitations are clearly described.	The research question is present but could be more focused. The data is described, but limitations are not fully explored.	The research question is "unclear" or poorly defined. The data description is underdeveloped.	Fails to define a research question or describe the data.
4. Methodology & Analysis	"Cadets effectively utilized tools... and explored novel data science methodologies". Explanation of novel or complex methods are clear and articulate	"Cadets utilized tools... and explored novel data science methodologies... but could have further developed the integration". Explanations are present but may lack some clarity.	"Cadets utilized tools... but did not explore novel data science methodologies". The description of progress is limited.	"Cadets utilized few tools... and did not explore novel data science methodologies". "Progress is minimal or incomplete".
5. Results & Discussion	Provides a "comprehensive" discussion of outputs and models. Fully "embrace[s] failed models and results".	Discusses results, but "could have been more concise or focused". "Failures are mentioned but could be analyzed in greater depth".	"Briefings were underdeveloped or lacked focus, missing important context". "Fails to discuss failed models" or lessons learned.	"Briefings were poorly prepared or incomplete". There is "little to no discussion of results" or lessons learned.
6. Conclusions	Draws clear, data-driven conclusions that directly address the research questions and sponsor's needs. Provides actionable recommendations	Conclusions are present, but the connection to the data, the research question, or the sponsor's problem could be more clearly articulated.	Conclusions are vague or do not logically follow from the results presented.	No clear conclusions or recommendations are provided based on the work.
7. Technical Communication & Professionalism	"Briefings were well-prepared, clear, and comprehensive". "Verbal... communication was cogent, accurate, and formal". All members speak and contribute professionally.	"Briefings were generally well-prepared". Communication is "mostly accurate and formal, with some areas needing improvement". All members speak.	"Briefings were underdeveloped". "Communication... was sometimes unclear or unprofessional". "Not all members speak".	"Briefings were poorly prepared or incomplete". "Verbal... communication was often lacking professionalism, coherence, and clarity". (F) Briefing is "absent".